

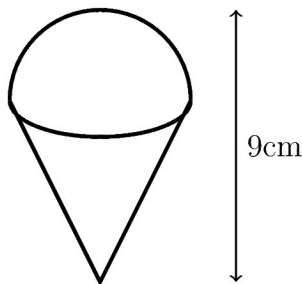
SAT SOLIDS (Area and Volume)

1. [March US 2023]

A right circular cylinder has a height of 6 inches. The radius of the base of the cylinder is 5 inches. What is the volume, in cubic inches, of the cylinder?

- A) 10π
- B) 30π
- C) 50π
- D) 150π

2. [March US 2023]



A hemisphere and a cone have circular bases of equal circumference and have been connected at their bases, as shown in the figure. The diameter of the base of the hemisphere is 6 centimeters(cm) and the total height of the figure is 9 cm. What is the total volume, in cm^3 , of the cone and hemisphere

- A) 18π
- B) 36π
- C) 45π
- D) 162π

3. [June 2023]

The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x+6$ centimeters, and the height of the triangle is $x-2$ centimeters. What is the values of x ?

4. [June 2023]

A right circular cylinder has a height of 4 meters(m) and a base with a radius of 18m. What is the volume, in m^3 , of cylinder?

- A) 4π
- B) 22π
- C) 288π
- D) 1296π

5. [June 2023]

The volume of a right rectangular prism with a square base is 2.448 cubic centimeters. If the area of the square base is 144 square centimeters, what is the area, in square centimeters, of one of the four lateral faces of the prism?

- A) 17
- B) 204
- C) 540
- D) 816

6. [November 2023]

The volume of right rectangular prism with a square base is 1,792 cubic centimeters. If the area of the square base is 64 square centimeters, what is the area, in square centimeters, one of the four lateral faces of the prism?

- A) 28
- B) 224
- C) 416
- D) 896

7. [December 2023]

The area of a rectangle is 630 square inches. The length of the rectangle is 90 inches. What is the width, in inches, of this rectangle?

- A) 7
- B) 90
- C) 315
- D) 540

8. [March 2024]

A hemisphere is half of a sphere. If a hemisphere has a radius of 50 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

- A) 5,200
- B) 20,900
- C) 196,300
- D) 261,800

9. [December 2023]

The table gives the areas and perimeters of two similar rectangles, where n is a constant.

	Area (square inches)	Perimeter (inches)
Rectangle A	2,450	490
Rectangle B	9,800	n

What is the value of n ?

- A) 980 C) 3,920
B) 1,960 D) 4,900

10. [March 2024]

To study fluctuations in leaf water potential, samples of wood were taken from 25 trees and cut in the shape of a cube. The length of the edge of one of these cubes is 2.00 centimeters. If this cube has a mass of 2.72 grams, what is the density of this cube, in grams per cubic centimeter?

11. [March 2024]

The length of the diagonal of a square is $\frac{148\sqrt{2}}{2}$ units. What is the area, in square units, of the square?

- A) 74 C) 5,476
B) 296 D) 10,952

12. [March 2024]

The floor of a school has an area of 2,025 square meters. An architect creates a scale model of the floor of the school, where the length of each side of the model is $\frac{1}{15}$ times the length of the corresponding side of the actual floor of the school. What is the area, in square meters, of the scale model?

- A) 9 C) 135
B) 15 D) 225

13. [March 2024]

	Volume (cubic units)
Right circular cylinder A	112π
Right circular cylinder B	$3,024\pi$

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B. The radius of right circular cylinder A is 4 units. The surface area of right circular cylinder A is $k\pi$ square units, and the surface area of right circular cylinder B is $n\pi$ square units, where k and n are constants. What is the value of $n - k$? (The surface area of a right circular cylinder with radius r and height h is $2\pi r^2 + 2\pi rh$.)

14. [March 2024]

$$\frac{4}{7}(4x + 7)(x + \sqrt{4k + 7})(x - \sqrt{4k + 7}) = 0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 77. What is the value of k ?

15. [March 2024]

Each side of square A has a length of 11 inches. Each side of square A is multiplied by a scale factor of 3 to create square B . What is the length, in inches, of each side of square B ?

- A) 8 C) 14
B) 11 D) 33

16. [March 2024]

The width of a rectangle is 7 centimeters. The length of the rectangle is 40 centimeters longer than the width. What is the area, in square centimeters, of this rectangle?

- A) 7 C) 54
B) 14 D) 329

17. [March 2024]

A polygon has 16 sides has a perimeter of 126. The length of one side is 6 inches. The other 15 sides have equal lengths. What is the length, in inches, of one of the 15 sides with equal lengths?

18. [March 2024]

The length of the diagonal of a square is $\frac{158\sqrt{2}}{2}$ units. What is the area, in square units, of the square?

- A) 79
- B) 316
- C) 6241
- D) 12482

19. [March 2024]

The floor of a school has an area of 4400 square meters. An architect creates a scale model of the floor of the school, where the length of each side of the model is $\frac{1}{20}$ times the length of the corresponding sides of the actual floor of the school. What is the area, in square meters, of the scale model?

- A) 11
- B) 20
- C) 220
- D) 275

20. [May US 2024]

A hemisphere is half of a sphere. If a hemisphere has a radius of 20 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

- A) 800
- B) 3,400
- C) 12,600
- D) 16,800

21. [May 2024]

A sphere has a radius of $\frac{13}{5}$ feet. What is the volume, in cubic feet, of the sphere?

- A) $\frac{8,788\pi}{375}$
- B) $\frac{28\pi}{5}$
- C) $\frac{52\pi}{15}$
- D) $\frac{5\pi}{13}$

22. [May 2024]

A cube has an edge length of 82 inches. A solid sphere with a radius of 41 inches is inside the cube, such that the sphere touches the center of each face of the cube. To the nearest cubic inch, what is the volume of the space in the cube not taken up by the sphere?

- A) 262,672
- B) 288,532
- C) 334,846
- D) 546,087

23. [May 2024]

Rectangles $ABCD$ and $EFGH$ are similar. The length of each side of $EFGH$ is 6 times the length of the corresponding side of $ABCD$. The area of $ABCD$ is 96 square units. What is the area, in square units, of $EFGH$?

- A) 16
- B) 36
- C) 576
- D) 3,456

24. [May US 2024]

The study fluctuations in leaf water potential, samples of wood were taken from 24 trees and cut in the shape of a cube. The length of the edge of one of those cubes is 2.00 centimeters. If this cube has a mass of 3.04 grams, what is the density of this cube, in grams per cubic centimeter?

25. [May 2024]

A circle is inscribed in a square such that the circumference of the circle touches the midpoint of each side of the square. The length of the diagonal of the square is 176 units. What is the area, in square units, of the circle?

- A) $3,872\pi$
- B) $7,744\pi$
- C) $15,488\pi$
- D) $30,976\pi$

26. [May 2024]

Rectangle $ABCD$ is similar to rectangle $EFGH$. The area of rectangle $ABCD$ is 648 square inches, and the area of rectangle $EFGH$ is 72 square inches. The length of the longest side of rectangle $ABCD$ is 36 inches. What is the length, in inches, of the longest side of rectangle $EFGH$?

- A) 4 C) 12
B) 9 D) 36

27. [June 2024]

A chemist combines water and isopropanol to make a mixture with a volume of **52 milliliters(mL)**. The volume of isopropanol in the mixture is **10mL**. What is the volume of water, in **mL**, in the mixture?(Assume that the volume of the mixture is the sum of the volumes of water and isopropanol before they were mixed.)

28. [June 2024]

A rectangular banner has an area of 2,500 square inches. A copy of the banner is made in which the length and width of the original banner are each increased by 40%. What is the area of the copy, in. square inches?

- A) 2,540 C) 3,500
B) 2,580 D) 4,900

29. [June 2024]

A chemist combines water and isopropanol to make a mixture with a volume of **45 milliliters (mL)**. The volume of isopropanol in the mixture is **10 mL**. What is the volume of water, in **mL**, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and isopropanol before they were mixed.)

30. [June 2024]

A cube has a volume of 79,507 cubic units. What is the surface area, in square units, of the cube?

31. [June 2024]

A cylinder has a diameter of 14 inches and a height of 8 inches. What is the volume, in cubic inches, of the cylinder?

- A) 49π C) 392π
B) 112π D) $1,568\pi$

32. [August 2024]

A right rectangular prism has a length of 26 centimeters(cm), a width of 16cm, and a height of 14cm. What is the surface area, in cm^2 , of the right rectangular prism?

33. [August 2024]

Rectangle $ABCD$ and $EFGH$ are similar. The length of each side of $EFGH$ is 10 times the length of the corresponding side of $ABCD$. The area of $ABCD$ is 100 square units. What is the area, in square units, of $EFGH$?

- A) 10 C) 1,000
B) 100 D) 10,000

34. [August 2024]

A right circular cylinder has a radius of 12 centimeters and a volume of $2,448\pi$ cubic centimeters. What is the height, in centimeters, of the cylinder?

35. [August US 2024]

Square P has a side length of x inches. Square Q has a perimeter that is 32 inches greater than the perimeter of square P . The function f gives the area of square Q , in square inches. Which of the following defines f ?

- A) $f(x) = (x + 8)^2$
B) $f(x) = (x + 32)^2$
C) $f(x) = (32x + 8)^2$
D) $f(x) = (32x + 32)^2$

36. [August 2024]

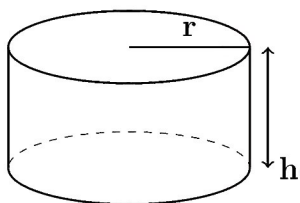
The floor of a library has an area of 9375 square feet. An architect creates a scale model of the floor of the library, where the length of each side of the model is $\frac{1}{25}$ times the length of the corresponding side of the floor of the actual library. What is the area, in square feet, of the scale model?

37. [August 2024]

A right square pyramid has a total surface area of 36,864 square inches, and the combined surface area of the four lateral faces of this pyramid is 20,480 square inches. What is the height, in inches, of this pyramid?

- A) 48 C) 80
B) 64 D) 128

38. [August US 2024]



The figure shown is a right circular cylinder with a radius r and a height of h . A second right circular cylinder (not shown) has a volume that is 576 times as large as the volume of the cylinder shown. Which of the following could represent the radius R , in terms of r , and the height H , in terms of h , of the second cylinder?

- A) $R = 9r$ and $H = 8h$
B) $R = 9r$ and $H = 64h$
C) $R = 8r$ and $H = 9h$
D) $R = 64r$ and $H = 9h$

39. [August US 2024]

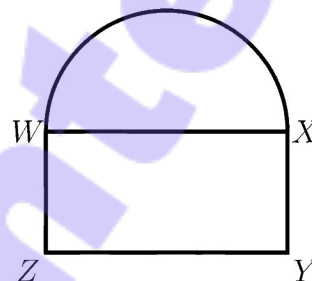
A right circular cone has a volume of $2,700\pi$ cubic centimeters and the area of its base is 225π square centimeters. What is the slant height, in centimeters, of this cone?

- A) 12 C) 36
B) 15 D) 39

40. [August US 2024]

The length of each side of a box is 94 centimeter. Each side of the box is in the shape of a square. The box does not have a lid. What is the exterior surface area, in square meters, of this box without a lid? (1 meter = 100 centimeters)

41. [October US 2024]



Note: Figure not drawn to scale

The figure shown consists of a rectangle and a semicircle, where the length of $\overline{WX}$ is 20 units and the length of $\overline{WZ}$ is 5 units. The diameter of the semicircle is $\overline{WX}$. Which of the following expressions represents the area, in square units, of the figure?

- A) $20 + 5 + 10\pi$ C) $20 \times 5 + 50\pi$
B) $20 \times 5 + 10\pi r$ D) $20 \times 5 + 100\pi$

42. [October US 2024]

To study fluctuations in leaf water potential, samples of wood were taken from 27 trees and cut in the shape of a cube. The length of the edge of one of these cubes is 3.000 centimeters. This cube has a density of 0.170 grams per cubic centimeter. What is the mass of this cube, in grams?

43. [October US 2024]

The volume of a right rectangular prism with a square base is 539 cubic centimeters. If the area of each of the four lateral faces is 77 square centimeters, what is the height, in centimeters, of the prism?

- A) 77 C) 11
B) 49 D) 7

44. [October 2024]

To study fluctuations in leaf water potential, samples of wood were taken from 22 trees and cut in the shape of a cube. The length of the edge of one of these cubes is 3.000 centimeters. This cube has a density of 0.250 grams per cubic centimeter. What is the mass of this cube, in grams?

45. [October 2024]

A right triangular pyramid has exactly six edges. Each of these edges is 96 centimeters long. If the surface area of the pyramid is $k\sqrt{3}$ square centimeters, what is the value of k ?

46. [October US 2024]

The length of square M is 9 times bigger than one side of square N . The area of the square N is 25 square centimeters. What is the area, in square centimeters, of square M ?

- A) 2025 C) 405
B) 810 D) 225

47. [October US 2024]

The height of a right-angular cylinder is 6 inches longer than its radius, and the surface are of the cylinder is 1080π square inches. What is the cylinder's radius, in inches?

- A) 15 C) 21
B) 18 D) 24

48. [October US 2024]

To study fluctuations in the leaf water potential, samples of wood were taken from 26 trees and cut in the shape of a cube. The length of the edge of one of these cubes is 2.00 centimeters. If this cube has a mass of 5.12 grams, what is the density of this cube in grams per cubic centimeter?

49. [October US 2024]

A right triangular prism has a base of $28p$ square centimeters((cm^2)). The length of the rectangular prism is 12 cm, and the height of the rectangular prism is 4 cm. Which expression represents the surface area, in cm^2 , of the right rectangular prism?

- A) $\frac{512p}{3}$
A) $\frac{224p}{3} + 96$
A) $56p + 192$
A) $560p + 96$

50. [October US 2024]

Rectangle $ABCD$ and $EFGH$ are similar. The length of each side of $EFGH$ is 6 times the length of the corresponding side of $ABCD$. The ares of $ABCD$ is 102 square units . What is the area, in square units, of $EFGH$?

- A) 17 C) 612
B) 36 D) 3672

51. [October US 2024]

A hemisphere is half of a sphere. If a hemisphere has a radius of 70 inches, which of the following is closest to the volume of this hemisphere?

- A) 10300 C) 538800
B) 41100 D) 718400

52. [November 2024]

A right square pyramid has a height of 11 centimeters (cm). A second right square pyramid has a height of 22 centimeters. The are of the base of each of the two pyramids is 100 cm^2 . Which of the following is closest to the difference in the surface area of the second pyramid and the surface area of the first pyramid, in cm^2 ?

- A) 209 D) 367
B) 220 C) 551

53. [December US 2024]

The circumference of the base of a right circular cylinder is 6π meters, and the height of the cylinder is 19 meters. What is the volume, in cubic meters, of the cylinder?

- A) 57π C) 171π
B) 114π D) 684π

54. [December 2024]

A right circular cone has a height of 13 centimeters (cm) and a base with a radius of 6 cm. What is the volume, in cm^3 , of this cone?

- A) 26π C) 156π
B) 78π D) 468π

55. [December 2024]

A sphere has a radius of $\frac{8}{3}$ feet. What is the volume, in cubic feet, of the sphere?

- A) $\frac{3\pi}{8}$ C) $\frac{17\pi}{3}$
B) $\frac{32\pi}{9}$ D) $\frac{2018\pi}{81}$